

Chapter 8:

Integration Techniques and Improper Integrals

Section 8.1: Basic Integration Rules

REVIEW OF BASIC INTEGRATION RULES ($a > 0$)

1. $\int kf(u)\,du = k\int f(u)\,du$
2. $\int [f(u) \pm g(u)]\,du = \int f(u)\,du \pm \int g(u)\,du$
3. $\int du = u + C$
4. $\int u^n\,du = \frac{u^{n+1}}{n+1} + C, \quad n \neq -1$
5. $\int \frac{du}{u} = \ln|u| + C$
6. $\int e^u\,du = e^u + C$
7. $\int a^u\,du = \left(\frac{1}{\ln a}\right)a^u + C$
8. $\int \sin u\,du = -\cos u + C$
9. $\int \cos u\,du = \sin u + C$
10. $\int \tan u\,du = -\ln|\cos u| + C$
11. $\int \cot u\,du = \ln|\sin u| + C$
12. $\int \sec u\,du = \ln|\sec u + \tan u| + C$
13. $\int \csc u\,du = -\ln|\csc u + \cot u| + C$
14. $\int \sec^2 u\,du = \tan u + C$
15. $\int \csc^2 u\,du = -\cot u + C$
16. $\int \sec u \tan u\,du = \sec u + C$
17. $\int \csc u \cot u\,du = -\csc u + C$
18. $\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin\frac{u}{a} + C$
19. $\int \frac{du}{a^2 + u^2} = \frac{1}{a}\arctan\frac{u}{a} + C$
20. $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a}\operatorname{arcsec}\frac{|u|}{a} + C$

19. $\displaystyle\int \frac{du}{a^2 + u^2} = \frac{1}{a}\arctan\frac{u}{a} + C$

EXAMPLE

Find each integral.

a. $\displaystyle\int \frac{4}{x^2 + 9}\,dx$ **b.** $\displaystyle\int \frac{4x}{x^2 + 9}\,dx$ **c.** $\displaystyle\int \frac{4x^2}{x^2 + 9}\,dx$

SOLUTION **a.** Use the Arctangent Rule and let $u = x$ and $a = 3$.

$$\int \frac{4}{x^2 + 9}\,dx = 4\int \frac{1}{x^2 + 3^2}\,dx$$ Constant Multiple Rule

$$= 4\left(\frac{1}{3}\arctan\frac{x}{3}\right) + C$$ Arctangent Rule

$$= \frac{4}{3}\arctan\frac{x}{3} + C$$ Simplify.

EXAMPLE

Find each integral.

a. $\int \frac{4}{x^2 + 9}\,dx$ **b.** $\int \frac{4x}{x^2 + 9}\,dx$ **c.** $\int \frac{4x^2}{x^2 + 9}\,dx$

SOLUTION

b. The Arctangent Rule does not apply because the numerator contains a factor of x. Consider the Log Rule and let $u = x^2 + 9$. Then $du = 2x\,dx$, and you have

$$\int \frac{4x}{x^2 + 9}\,dx = 2\int \frac{2x\,dx}{x^2 + 9}$$ Constant Multiple Rule

$$= 2\int \frac{du}{u}$$ Substitute: $u = x^2 + 9$.

$$= 2\ln|u| + C$$ Log Rule

$$= 2\ln(x^2 + 9) + C.$$ Rewrite as a function of x.

EXAMPLE

Find each integral.

a. $\int \frac{4}{x^2 + 9}\,dx$ **b.** $\int \frac{4x}{x^2 + 9}\,dx$ **c.** $\int \frac{4x^2}{x^2 + 9}\,dx$

SOLUTION

c. Because the degree of the numerator is equal to the degree of the denominator, you should first use division to rewrite the improper rational function as the sum of a polynomial and a proper rational function.

$$\int \frac{4x^2}{x^2 + 9}\,dx = \int \left(4 + \frac{-36}{x^2 + 9}\right) dx$$ Rewrite using long division.

$$= \int 4\,dx - 36\int \frac{1}{x^2 + 9}\,dx$$ Rewrite as two integrals.

$$= 4x - 36\left(\frac{1}{3}\arctan\frac{x}{3}\right) + C$$ Integrate.

$$= 4x - 12\arctan\frac{x}{3} + C$$ Simplify.

EXAMPLE

Find $\int \frac{4x^3}{x^2+9}\,dx$

SOLUTION

$$I = 2x^2 - 18\ln(x^2 + 9) + C$$

EXAMPLE

Find $\int \frac{4x^4}{x^2+9}\,dx$

SOLUTION

$$I = \frac{4}{3}x^3 - 36x + 108\arctan\left(\frac{x}{3}\right) + C$$

EXAMPLE

Find $\int \frac{4x^5}{x^2+9}\,dx$

SOLUTION

$$I = x^4 - 18x^2 - 162\ln(x^2 + 9) + C$$

EXAMPLE A Substitution Involving $a^2 - u^2$

Find $\int \frac{x^2}{\sqrt{16 - x^6}}\, dx.$

SOLUTION

$$I = \frac{1}{3}\arcsin\left(\frac{x^3}{4}\right) + C$$

EXAMPLE A Disguised Form of the Log Rule

Find $\int \frac{1}{1 + e^x}\, dx.$

SOLUTION

$$I = x - \ln(1 + e^x) + C$$

EXAMPLE A Disguised Form of the Power Rule

Find $\int (\cot x)[\ln(\sin x)]\,dx.$

SOLUTION

$$I = \frac{1}{2}\,(\ln(\sin x))^2 + C$$

EXAMPLE Using Trigonometric Identities

Find $\int \tan^2 2x \, dx$.

SOLUTION

$$I = \frac{1}{2}\tan 2x - x + C$$

Solve exercises of section 8.1

Start with the suggested problems mentioned in the syllabus.